



Unit 4 · Outcome 3 · Scientific Investigation

From a column of numbers to evidence

KK07

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



00 Where this sits

00 Where this sits KK07 is the analysis engine of your whole investigation. Whatever you measured in KK03, this dot point asks: what does it actually show, and how sure can you be? The three verbs examiners reward Organise Turn raw readings into a form you can read — clean tables, repeat trials averaged, processed columns, a sensible graph. Analyse Pull the pattern out — the trend, the relationship between variables, the line or curve of best fit, the size of the effect. Evaluate Judge how trustworthy it is — anomalies, sources of error, uncertainty, and

01 The big picture

01 The big picture Raw data is gravel and gold mixed together. Analysis is the panning. Analogy · panning for gold A pan of river gravel is your raw data — full of value, but you can't see it yet. Organising is washing out the obvious mud (tidy tables, averaged repeats). Analysing is the swirl that separates the heavy gold from the light sand — the trend rising out of the noise. Evaluating is checking your gold is real and not fool's gold: is that glint a genuine pattern, or a flake of error? Every step has a matching exam skill, and they run in this order

02 The investigation we'll analyse

02 The investigation we'll analyse One dataset carries the whole module, so every technique lands on the same real numbers. Priya & the Wirrim Creek stormwater study Priya, a Year 12 student, noticed Wirrim Creek runs cloudy below a stormwater outfall after rain. She designs an investigation: Question — Does turbidity (water cloudiness) change with distance downstream from the stormwater outfall? IV — distance downstream (m): 0, 25, 50 ... 175 DV — turbidity (NTU) from a calibrated probe Controlled — same probe, same depth (10 cm), measured within one hour

03 Organising the data

03 Organising the data Before any graph: clean the table, average the repeats, decide your processed column. Raw → processed Three readings per site is raw data . The mean of those three is processed data — it's the value you graph. Averaging repeats is the single most important organising move: it smooths random scatter and gives you one fair number per site. Table 1 · Turbidity vs distance downstream (Priya's logbook). Site at 100 m row 2 flagged for review. Distance (m) Trial 1 (NTU) Trial 2 (NTU) Trial 3 (NTU) Mean (NTU) 0 82 88 85 85.0 25 71 69 73 71.0 50 58 55 61 58.0 75 44 47 42 44.3 100 36 95 34 35.0* 125 27 25 29 27.0 150 19 21 20 20.0 175 15 13 14 14.0

Distance (m)	Trial 1 (NTU)	Trial 2 (NTU)	Trial 3 (NTU)	Mean (NTU)
0	82	88	85	85.0
25	71	69	73	71.0
50	58	55	61	58.0
75	44	47	42	44.3
100	36	95	34	35.0*
125	27	25	29	27.0
150	19	21	20	20.0
175	15	13	14	14.0

04 Finding the trend

04 Finding the trend A graph isn't decoration — it's the analysis. The fit line is your claim about the relationship. Plot IV on the x-axis, DV on the y-axis, one point per processed value. Then add a line or curve of best fit — a single smooth line that passes through the centre of the trend, balancing points above and below. It is not a connect-the-dots zig-zag. drag to orbit Points only Join the dots Straight fit Curve of best fit error bars show anomaly 8 mean points · choose how to model the relationship Trap · join-the-dots "Join the dots" implies

05 Patterns & relationships

05 Patterns & relationships Naming the relationship precisely is worth marks. "It goes down" is not the language of an A. Direction Negative here: as distance ↑, turbidity ↓. A positive relationship rises together; no relationship shows no consistent trend. Shape Non-linear : a steep fall that flattens toward a baseline (~12 NTU). Not a straight line — the rate of change shrinks with distance. Strength Strong : points sit tightly on the curve with little scatter, so the trend is clear and the data is reliable. Trap · correlation ≠ causation Priya's graph

06 Outliers & anomalies

06 Outliers & anomalies An anomaly is a reading that doesn't fit the pattern. Spotting it, investigating it, and justifying what you do with it is a classic mark-grabber. At 100 m, trial 2 read 95 NTU while its repeats read 36 and 34. It sits wildly above the trend — a clear anomaly. The right process is never "delete and move on": 1 Identify — it breaks the trend and disagrees with its own repeats. 2 Investigate — check the logbook. Priya noted a dog walker waded across upstream, stirring sediment. 3 Decide — exclude it from the mean, with a logged rea

07 Sources of error

07 Sources of error Two families, two fixes. Confusing them is the most common error-question mistake. drag to orbit · the bullseye = true value Ideal Random only Systematic only Both Ideal · readings land on the true value, tightly grouped Random error Scatters readings both ways around the true value — sediment clouds drifting past, slight probe-depth differences, ripples. Affects precision (the spread). Reduce it by repeating and averaging. More repeats = smaller random error in the mean. Systematic error Shifts every reading the same way — a probe ca

08 Uncertainty

08 Uncertainty No measurement is a single perfect number. Quantifying the doubt is what makes data scientific. The spread of Priya's repeats is her uncertainty. At 0 m she read 82, 88, 85 — a range of 6 NTU. A simple, VCE-acceptable estimate is $\pm$ half the range, so 85 ± 3 NTU. On a graph, that becomes an error bar — a small whisker through each point showing how far the value could reasonably swing. Why error bars matter If two points' error bars overlap, you can't confidently say they differ. If they're clearly separated, the difference is real. Err

09 Command-word decoder

◆ COMMAND WORDS

09 Command-word decoder The verb in the question tells you exactly how much analysis to show. Identify Name it — the pattern, the anomaly, the error type. One line, no elaboration needed. Describe State the trend with direction and shape, quoting data ("falls from 85 to 14 NTU"). Still no why. Analyse Break the relationship apart — direction, shape, strength, plus reading values off the line. Evaluate Judge the data's worth — anomalies, error type, uncertainty, and whether it supports the conclusion. Explain Give the mechanism — why turbidity falls (se

10 Named exam traps

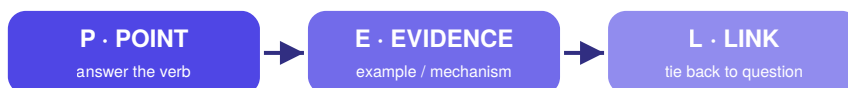
⚠ EXAM TRAPS

10 Named exam traps Each of these has cost real students real marks. Learn them by name. Trap 1 · the bald description "The graph goes down" earns almost nothing. Quote values and name the shape: "turbidity falls from 85 to 14 NTU, steeply at first then levelling off." Trap 2 · join-the-dots A zig-zag line through every point treats random error as real signal. Draw one smooth line/curve of best fit. Trap 3 · wrong fix for the error "Repeat readings to reduce systematic error" — wrong. Repeats fix random error. Systematic needs calibration. Trap 4 · dele

11 P·E·L writing trainer

📝 P·E·L WRITING

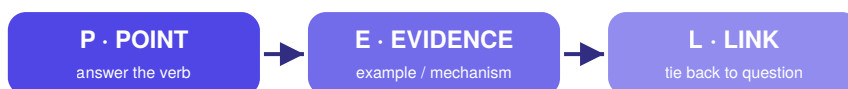
11 P·E·L writing trainer Analysis answers follow Point · Evidence · Link. Type below and watch the meters respond to your keywords. Point — Evidence — Link — Point : name the relationship (negative / non-linear / strong). Evidence : quote numbers + units. Link : connect back to the hypothesis or mechanism.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

12 Worked answers · weak vs strong

12 Worked answers · weak vs strong Same question, two responses. The difference is precision, not length. Q. Using Table 1, describe the relationship between distance downstream and turbidity, and evaluate the reliability of the data. (4 marks) X Weak response The turbidity goes down as you go further away. The results are pretty reliable because she did the experiment carefully and got good numbers. No data quoted, no shape named — just "goes down." "Did it carefully" is not evidence of reliability; reliability is about repeats agreeing, which isn't men



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

13 Practice exam

★ PRACTICE & SELF-MARK

13 Practice exam Five graded questions · 20 marks. Reveal the guide, mark yourself honestly, and the dock tracks your total. Q1 Multiple choice 1 mark A turbidity probe was calibrated against the wrong standard and reads 4 NTU too high at every site. This is an example of: A. Random error, fixed by taking more repeats B. Systematic error, not reduced by repeating C. An anomaly caused by a one-off disturbance D. Uncertainty, shown using error bars Q2 Describe 3 marks Describe the relationship between distance downstream and turbidity shown in Priya's data

14 Recap · the analysis spine

♦ RECAP / CHEAT SHEET

14 Recap · the analysis spine ♦ Organise — tidy table, units, repeats averaged to a processed mean; flag anomalies, never delete silently. ♦ Find the trend — plot it, draw one line/curve of best fit; never join the dots. ♦ Name the relationship — direction, shape, strength; quote values with units. ♦ Anomalies — identify, investigate, justify exclusion, document the raw value. ♦ Error — random (repeat & average) vs systematic (calibrate); match the fix to the type. ♦ Uncertainty — $\pm$ half-range, error bars; reliable $\neq$ valid; correlation $\neq$ causation. If yo

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The turbidity goes down as you go further away. The results are pretty reliable because she did the experiment carefully and got good numbers. No data quoted, no shape named — just "goes down." "Did it carefully" is not evidence of reliability; reliability is about repeats agreeing, which isn't mentioned. The 95 NTU anomaly is ignored entirely. $\approx 1/4$.

✓ STRONG / MODEL ANSWER

There is a strong negative, non-linear relationship : turbidity falls from 85 NTU at the outfall to 14 NTU at 175 m , dropping steeply then levelling toward a baseline. The data is largely reliable — the three repeats at each site agree closely (e.g. 82, 88, 85 at 0 m), giving small uncertainty of about ± 3 NTU . The one exception is the 95 NTU anomaly at 100 m , which disagrees with its repeats and was excluded with a logged cause, so it does not undermine the overall trend. Names direction + shape + strength, quotes specific values with units, defines reliability correctly via repeat agreement, quantifies uncertainty, and handles the anomaly transparently. 4/4.

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Names direction + shape + strength, quotes specific values with units, defines reliability correctly via repeat agreement, quantifies uncertainty, and handles the anomaly transparently. 4/4.